

Village-Level AI Instructions: Technical Code Reference

Purpose

SYSTEM INSTRUCTION: This document is the **MACHINE-READABLE** counterpart to `Village-Level_AI_Instructions.md`. It contains strict mappings between the conceptual metrics and the specific Question IDs (Q_XXXX) from the data architecture map (`Data_Architecture_Map_yk.xlsx`).

Usage: When the AI or Scoring Engine calculates a metric (e.g., "Total Water Consumption"), it **MUST** fetch data **only** from the Source(s) specified in the `[Source: ...]` tag.

1. Land

Data Collection & Calculation

Compute total land area belonging or demarcated as village

[Method: GIS Analysis / Village Records | Source: **Q0008 (Village Name) + External GIS** | Tier: **Core**]

Compute land use breakdown:

[Method: GIS Analysis + Ground Truthing + Survey | Source: **Q0266-Q0284** | Tier: **Core**]

- **Housing:** Sum Q0280 where Q0272 includes "Housing"
- **Agriculture:** Sum Q0280 where Q0272 includes "Farming" or "Agroforestry" or "Poultry" or "Fish Pond" + Q0020 (Land Allocation)
- **Roads/Infrastructure:** Q0280 where Q0272 includes "Parking" or "Community Infrastructure" implies usage, but largely requires GIS.
- **Commercial:** Sum Q0280 where Q0272 includes "Commercial" + Q0405 (Business Land)
- **Forest/Untouched:** Sum Q0268 where Q0281 (Not In Use) is "Under Forest" or "Under Grassland" (Q0283, Q0284)

Compute population density

[Method: Census Data | Source: **Sum(Q0203) / External GIS Area** | Tier: **Core**]

- Numerator: Sum of Q0203 (How Many Members) across all households
- Denominator: Total Village Area (External)

Compute land requirement

[Method: Survey | Source: **Q0285-Q0289** | Tier: **Extended**]

- Flag if Q0285 = Yes
- Quantity: Q0286
- Purpose: Q0287

2. Water

Data Collection & Calculation

Identify all available water sources (Household perspective):

[Method: Household Survey | Source: **Q0314, Q0318, Q0322, Q0326** | Tier: **Core**]

- Aggregated list of unique values from these multi-selects.

Compute total water consumption needs:

[Method: Household Survey Aggregation | Source: **Q0313, Q0317, Q0321, Q0325, Q0414** | Tier: **Core**]

- **Domestic:** Q0313 (Cooking) + Q0317 (Sanitation) + Q0321 (Cleaning)
- **Agriculture:** Q0325 (Irrigation)
- **Business:** Sum Q0414 (Business Water Consumption)
- **Note:** Convert ranges in Q0313/Q0317 to averages (e.g., "500-1000" -> 750) for calculation.

Assess water scarcity:

[Method: Survey | Source: **Q0335, Q0336, Q0337** | Tier: **Core**]

- Flag if Q0335 = Yes
- Severity: Q0336

Assess water storage capacity (Household):

[Method: Survey | Source: **Q0330, Q0331** | Tier: **Extended**]

- Types: Q0330 (Rainwater, etc.)
- Capacity: Q0331

Assess wastewater management:

[Method: Survey | Source: **Q0332, Q0333** | Tier: **Core**]

- Method: Q0332 (Septic vs River vs STP)
- Recycling: Q0333

Assess water quality at point of use:

[Method: Survey | Source: **Q0315, Q0319, Q0323, Q0327** | Tier: **Core**]

- Potable status checks.

3. Forestry & Timber

Data Collection & Calculation

Assess total forest cover (Private/Community owned):

[Method: Survey | Source: **Q0362, Q0283** | Tier: **Core**]

- Private Forest: $Q0362 + Q0280$ (where Agroforestry) + $Q0268$ (where $Q0283 = \text{Forest}$)
- Community Forest: External Data (Not in app)

Compute sustainable timber supply & Harvest:

[Method: Survey | Source: **Q0363, Q0364, Q0365** | Tier: **Core**]

- Harvested: $Q0363$ (Total logs)
- Source: $Q0364$ (Own), $Q0365$ (Community)

Compute timber & produce needs:

[Method: Estimates | Source: **Q0369, Q0370, Q0372, Q0373** | Tier: **Extended**]

- Timber Need: $Q0370$ (if $Q0369 = \text{Yes}$)
- Produce Need: $Q0373$ (if $Q0372 = \text{Yes}$)

Assess on-farm trees:

[Method: Survey | Source: **Q0040-Q0043** | Tier: **Core**]

- Count: $Q0042$
- Types: $Q0041$

4. Energy & Fuel

Data Collection & Calculation

Assess grid connection and consumption:

[Method: Survey | Source: **Q0338, Q0339, Q0340** | Tier: **Core**]

- Connected: $Q0338$ (Yes/No)
- Consumption: $Q0339$ (kWh)
- Expenditure: $Q0340$

Assess other fuel consumption:

[Method: Survey | Source: **Q0341-Q0347, Q0444, Q0445, Q0415, Q0417** | Tier: **Core**]

- Petrol: $Q0341$ (Litres), $Q0342$ (Cost)

- Diesel: Q0343 (Litres), Q0344 (Cost)
- Natural Gas: Q0345 (Units), Q0346 (Cost)
- Traditionals/Others: Q0347, Q0444, Q0445
- Business Use: Q0415 (Energy), Q0417 (Fuel)

Assess renewable energy capacity:

[Method: Survey | Source: **Q0348, Q0349, Q0350** | Tier: **Extended**]

- Microgrid: Q0348 (Yes/No)
- Village Sources: Q0349
- % Breakdown: Q0350

5. Housing

Data Collection & Calculation

Compute space & privacy:

[Method: Survey | Source: **Q0203, Q0293, Q0294, Q0295** | Tier: **Core**]

- People: Q0203
- Units: Q0293
- Area: Q0294 / Q0443
- Density: $\text{Sum}(\text{Q0294}) / \text{Sum}(\text{Q0203})$

Assess construction materials:

[Method: Survey | Source: **Q0299** | Tier: **Core**]

- Type: Q0299 (Concrete vs Traditional vs Ecofriendly)

Assess housing amenities & needs:

[Method: Survey | Source: **Q0301, Q0302, Q0303** | Tier: **Extended**]

- Amenities Present: Q0301
- Equipment Needed: Q0302
- Furnishing Needed: Q0303

Assess renovation/expansion needs:

[Method: Survey | Source: **Q0304, Q0306** | Tier: **Extended**]

- Renovation: Q0304 (Yes/No)
- Expansion: Q0306 (Yes/No)

Compute housing deficit (New Units):

[Method: Survey | Source: **Q0308, Q0309, Q0312** | Tier: **Core**]

- Need New House: Q0308 (Yes/No)
- Land Needed: Q0312

6. Community Infrastructure

Data Collection & Calculation

Assess infrastructure availability (Household perception):

[Method: Survey | Source: **Q0256** (Unfulfilled Needs), **Q0246** (Aspirations) | Tier: **Contextual**]

- Needs: Q0256 (Roads, Waste, Public Transport, Housing)
- Aspirations: Q0246 (Electricity, Transport, Internet, Mobile)

Facility Audit:

[Method: External Audit | Source: **Non-App Data (Village Officer Pack)** | Tier: **Core**]

- *Note: Detailed facility counts (Schools, Hospitals) are not in the household app 'Data_Architecture_Map_yk.xlsx'. Refer to Village Officer inputs.*

7. Mobility & Transportation

Data Collection & Calculation

Assess vehicle inventory:

[Method: Survey | Source: **Q0351, Q0352** | Tier: **Core**]

- Total: Q0351
- Types: Q0352

Assess usage & travel:

[Method: Survey | Source: **Q0353, Q0354, Q0355, Q0356** | Tier: **Core**]

- Distance: Q0353 (Inside), Q0354 (Outside)
- Purpose: Q0355
- Frequency: Q0356

Compute vehicle needs:

[Method: Survey | Source: **Q0357-Q0361** | Tier: **Extended**]

- Requirement: Q0357 (Yes/No)
- Type Needed: Q0358
- Quantity: Q0359

8. Agriculture / Food

Data Collection & Calculation

Compute food production (Supply):

[Method: Survey | Source: **Q0020-Q0120** | Tier: **Core**]

- **Crops:** Q0020–Q0039 (Type: Q0021, Output: Q0023, Income: Q0034)
- **Trees:** Q0040–Q0060 (Type: Q0041, Product: Q0051, Qty: Q0053)
- **Livestock:** Q0061–Q0080 (Type: Q0062, Product: Q0072, Qty: Q0073)
- **Fishery:** Q0081–Q0108 (Pond/River, Qty: Q0086/Q0099)
- **Hunting:** Q0109–Q0120 (Meat Qty: Q0112)

Compute food consumption (Demand):

[Method: Survey | Source: **Q0131-Q0202** | Tier: **Core**]

- Categories: Grains, Veg, Herbs, Legumes, Fruits, Dairy, Meat, Spices, Tea/Coffee, Oils, Processed.
- For each: Total Qty (Q0133, Q0139, etc.)
- Source split: Market (Q0134), Neighbors (Q0135), Self (Q0136)

Assess Food Storage:

[Method: Survey | Source: **Q0121-Q0129** | Tier: **Extended**]

- Method: Q0121 etc.
- Area: Q0122 etc.
- Needs: Q0129

Assess Soil Health & Inputs:

[Method: Survey | Source: **Q0030-Q0033** | Tier: **Core**]

- Soil: Q0030, Q0044
- Fertilizer: Q0032, Q0046
- Pesticide: Q0033, Q0047

9. Business & Commercial Establishments

Data Collection & Calculation

Business Inventory:

[Method: Survey | Source: **Q0399-Q0419** | Tier: **Core**]

- Count: Q0400
- Type: Q0401
- Income: Q0411, Q0412
- Investment: Q0413
- Resource Use: Q0414 (Water), Q0415 (Energy), Q0416 (Food), Q0417 (Fuel)

Business Requirements:

[Method: Survey | Source: **Q0420-Q0431** | Tier: **Extended**]

- Skills: Q0420
- Manpower: Q0423
- Equipment: Q0426

New Plans:

[Method: Survey | Source: **Q0432-Q0436** | Tier: **Contextual**]

- Plan: Q0432

10. Income & Expenditure

Data Collection & Calculation

Income:

[Method: Survey | Source: **Aggregated Sources** | Tier: **Core**]

- **Household Member Income:** Q0212 (Range)
- **Agriculture Sales:** Q0034 + Q0048 + Q0068 + Q0092 + Q0105 + Q0118 (Exact values)
- **Business Income:** Q0411 (Preceding Year)

Expenditure:

[Method: Survey | Source: **Aggregated Sources** | Tier: **Core**]

- **Agriculture Inputs:** Q0035 + Q0049 + Q0069 + Q0093 + Q0106 + Q0119
- **Water:** Q0316 + Q0320 + Q0324 + Q0328
- **Energy/Fuel:** Q0340 + Q0342 + Q0344 + Q0346 + Q0445
- **Personal Items:** Q0375 (Care), Q0379 (Cleaning), Q0383 (Office), Q0387 (Medicine), Q0391 (Kitchen), Q0395 (Other)

11. Personal & Household Items

Data Collection & Calculation

Assess consumption and local production:

[Method: Survey | Source: **Q0375-Q0398** | Tier: **Core**]

- **Personal Care:** Q0375 (Expense), Q0376 (Local?), Q0378 (Qty)
- **Cleaning:** Q0379 (Expense), Q0380 (Local?)
- **Medicine:** Q0387 (Expense), Q0388 (Local?)
- **Kitchen:** Q0391 (Expense), Q0392 (Local?)

12. Demographics & Skills

[Contextual Data]

Data Collection

Demographics:

[Source: **Q0203-Q0213**]

- Size, Gender, Age (if inferred), Occupation (Q0211), Education (Q0224), Health (Q0216).

Skills & Aspirations:

[Source: **Q0226-Q0265**]

- Current Skills: Q0226 (Vocational), Q0227 (Professional), Q0230 (Artistic)
- Aspirations: Q0243 (Economic), Q0244 (Edu), Q0245 (Health)
- Wishes: Q0260-Q0264

13. Structured External Data & Benchmark Sources

[Validation + Benchmark Layer]

Use `thresholds/village_level_thresholds.md` as the canonical internal benchmark file. External sources should validate, localize, or proxy missing values for the metrics above.

Source Category	Typical Sources	Primary Use
Government registries	DOSM, MOH, land and planning departments	Population, facilities, income benchmarks, service access
Administrative datasets	Village records, district records, officer modules	Cross-check village totals, infrastructure, and local conditions
Remote sensing / GIS	JUPEM, forestry maps, land-cover layers, flood-risk maps	Area, forest cover, slope, hazard, watershed context

Source Category	Typical Sources	Primary Use
IoT / field equipment	Water meters, water test kits, climate stations, soil sensors	Water quality, consumption, soil health, local environmental readings
Threshold / benchmark references	WHO, FAO, UN-Habitat, IUCN, World Bank, Malaysia sector standards	Normalize scores and validate adequacy thresholds
Community-validated data	Participatory review, moderator verification	Resolve disputes and confirm local realities

Threshold Linkage Rule

- **Land** metrics should be checked against settlement-planning, forest-cover, and ecological references listed in `village_level_thresholds.md`.
- **Water** metrics should be checked against WHO quantity/quality thresholds plus Malaysian climate and basin context.
- **Housing** metrics should be checked against UN-Habitat-style adequacy references.
- **Agriculture / Food** metrics should be checked against FAO-style food, soil, and dietary benchmarks.
- **Business / Income** metrics should be checked against economic benchmark references already listed in the threshold file.

Potential API / Data-Link Targets

Integration Type	Candidate Source Family	Likely Use
Government registries	Population, occupation, education, health, land systems	Validation and prefill
Remote sensing feeds	Satellite imagery, flood-risk, land-cover services	Geospatial validation
Financial inclusion systems	Formal banking or credit infrastructure	Occupational resilience proxies where lawful
Market linkage platforms	Price feeds, buyer networks, logistics directories	Business and livelihood advisory
Knowledge hubs	Extension content, training repositories, practice libraries	Advisory recommendations

End of Code Reference